WRITTEN TEST - INSTRUCTIONS	
Candidate Name:	
Assessor Name:	
Qualification:	**Certificate in AI in Immersive Technology, Level 6**
Unit of Competency	
Assessment Centre:	
Date of Assessment:	
Time of Assessment:	

Instructions:

Read and understand the directions carefully:

- this written examination is based on the performance criteria from all the units of competency in AI in Immersive Technology, Level 6.
- this assessment activity will be used to measure your underpinning knowledge
- write your answers on the paper provided
- answer all the questions as best as possible
- you have 1 (one) hour to complete this test

WRITTEN TEST

1 **What is the missing frequency when the arithmetic mean of the given data is 28?**

Profits per shop	0-10	10-20	20-30	30-40	40-50	50-60
No. of shops	12	18	27	?	17	6

2. **What is the variance of the following dataset: 5, 5, 5, 5**

3.	**What is the SD of a Bernoulli variable with p = 0.5?**
4.	**Consider the function: $f(x, y) = 3x^2 + 2xy + y^2$. Find the gradient of f(x, y).**
5.	**The mean number of a set of 5 numbers is 12.7. What extra number must be added to bring the mean up to 13.1?**
6.	**How do you define a variable and ensure it is treated as a float even if initialized with an integer?**

7.	**How do you print multiple variables with a custom separator?**
8.	**How do you skip the current iteration but continue the loop?**
9.	**How do you check if a string starts with a specific prefix?**
10.	**Define "Completeness" in a search algorithm.**

11.	**Explain the difference between Normalization and Standardization.**
12.	**Explain the "Ordinary Least Squares" (OLS) method.**
13.	**What is the difference between Feature Selection and Feature Extraction?**
14.	**What is a Chatbot in the context of NLP?**

15.	**How does Heuristic Search improve upon traditional search algorithms?**
15	**How does a Multi-Layer Perceptron (MLP) differ from a Single-Layer Perceptron?**
17	**How does "Term Frequency" (TF) contribute to feature importance?**
18	**What is the fundamental architecture of an Artificial Neural Network?**

19	**What is the "Feed-Forward Network" (FFN) within a Transformer block?**
20	**Which programming language is commonly used for Unity development?**
21	**Differentiate inside-out and outside-in tracking.**
22	**What is the difference between navigation and manipulation interfaces?**
23	**What is cybersickness, and what causes it?**

Q24. What is the difference between mobile and high-end VR platforms?

Mobile VR uses smartphones or standalone headsets (e.g., Meta Quest) — affordable, portable, but limited processing power and graphics. **High-end VR** is tethered to a powerful PC or console (e.g., Valve Index) — delivers superior graphics, tracking, and performance but is expensive and less portable.

Q25. Explain the mobile AR platform briefly.

Mobile AR uses a smartphone or tablet's camera, GPS, and sensors to overlay digital content onto the real world. Platforms like **ARKit** (Apple) and **ARCore** (Google) enable this by detecting surfaces and tracking the environment in real time.

Q26. What is inertial full-body tracking?

It uses **IMU sensors** (accelerometers + gyroscopes) placed on multiple body parts to track full-body movement. The sensors measure acceleration and rotation to reconstruct body pose in real time — no cameras needed, making it portable but prone to drift over time.

Q27. What is the role of a graphics accelerator?

A graphics accelerator (GPU) handles the heavy parallel processing needed to render images, textures, and 3D graphics quickly. It offloads this work from the CPU, enabling real-time, high-quality visuals in games, VR, and simulations.

Q28. Differentiate co-located rendering from distributed virtual environments.

Co-located rendering means all users share the same physical space and the rendering is done locally on one system. **Distributed virtual environments** connect users across different physical locations over a network, with rendering handled on separate systems — like online multiplayer VR.

29. How do you configure Windows MR?
Connect the Windows MR headset to the PC, open **Mixed Reality Portal**, follow the setup wizard, set the boundary, and pair the controllers.

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30. What is Singular Value Decomposition (SVD)?
SVD is a matrix decomposition method where a matrix A is written as $A = U\Sigma V^T$. It is used for dimensionality reduction and data analysis.